FRONT-END DEVELOPMENT
HACKATHON – Level-2 Template (Interactivity Implementation)

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TITLE: Activity Tracker - Interactivity

GITHUB LINK:https://github.com/crushsanjay007-cpu/Sanjay.git

1. Objective

The objective of this project is to implement front-end interactivity that allows users to update activity status and view real-time progress using JavaScript. The application enables users to mark activities as completed and dynamically updates the UI without reloading the page.

2. Problem Focus

In Level-1, the interface was static and only displayed activities. In Level-2, interactivity is added by implementing event handling. When users click a checkbox, the activity status changes and progress updates dynamically. This demonstrates state changes and real-time UI updates.

3. Application Overview

The application is an Activity Tracker that displays a list of tasks. Users can mark activities as completed using checkboxes. The completed activities are visually highlighted and the progress counter updates automatically.

4. Data Handling Approach

The application uses a JavaScript array to store activity data. Each activity contains a name and completion status. When a user marks an activity, the array is updated and UI re-rendered.

5. UI Components Design

The UI consists of:

Activity List

Checkbox Controls

Progress Display

These components are created using HTML and styled using CSS

6. Interactivity Implementation

Interactivity is implemented using checkbox event handling. When user clicks checkbox, activity status changes from pending to completed and UI updates.

JavaScript coding .

```
function toggleStatus(index) {
 activities[index].completed = !activities[index].completed;
 renderActivities();
}
```

7. Application Logic

The application counts completed activities and updates progress dynamically. The UI refreshes without page reload.

Java script coding.

```
function updateProgress() {
 let completed = activities.filter(a => a.completed).length;
 progress.innerText =
 `${completed} out of ${activities.length} completed`;
}
```

8. Technology Used

The technologies used in this project are HTML for structure, CSS for

styling, and JavaScript for interactivity and dynamic UI updates.

9. Best Practices Followed

Meaningful variable names are used. Functions are separated properly. Hardcoding is avoided. Comments are added. Edge cases like all tasks completed are handled.

10. Key Features Implemented

Mark activity as completed

Real-time progress update

Dynamic UI update

No page reload

11. Expected Output / Deliverables

codepen.io/pen

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Save | Settings

HTML

```html
    <p id="progress">0 out of 0 completed</p>

    <ul id="activityList"></ul>
</div>

<script src="script.js">
</script>
</body>
</html>
```

CSS

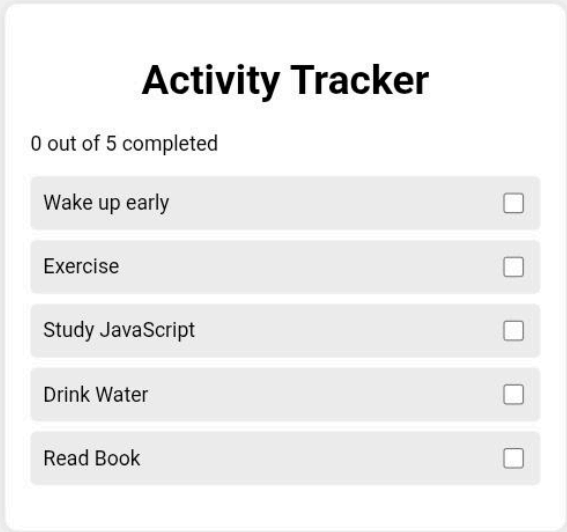
```css
    display: flex;
    justify-content: space-between;
    align-items: center;
    border-radius: 5px;
}

.completed {
    text-decoration: line-through;
    background: #c8f7c5;
}
```

JS

```js
    let completed = activities.filter(a => a.completed).length;
    progress.innerText = `${completed} out of ${activities.length} completed`;
}

// Initial Load
renderActivities();
```



12. Learning Outcome

This project helped understand JavaScript event handling, DOM manipulation, dynamic updates, and interactive UI development.

13. Conclusion

The application successfully implements interactivity using JavaScript.

Users can mark tasks as completed and view real-time progress, improving usability and user experience.